

DIGITAL SUMS OF POWERS, FACTORIALS, AND LEAST COMMON MULTIPLES

DAVID G. RADCLIFFE

ABSTRACT. We survey several lower bounds for digital sums. Elementary divisibility arguments give logarithmic lower bounds for many powers, factorials, and least common multiples. We then recall Stewart's theorem, which uses lower bounds for linear forms in logarithms to show that, whenever $\log a / \log b$ is irrational, the number of nonzero base- b digits of a^n tends to infinity, with an effective quantitative bound.

1. INTRODUCTION

Sierpiński asked for an elementary proof that the sum of the decimal digits of 2^n tends to infinity with n [6, Problem 209]. The problem is deceptive: unboundedness is not enough, since digital sums are highly nonmonotone, and even for powers such as 2^n the expected linear order of growth remains far beyond present methods.

This note treats the same phenomenon in a broader setting. Let $s_b(N)$ denote the sum of the base- b digits of N , and let $c_b(N)$ denote the number of nonzero base- b digits. Since

$$c_b(N) \leq s_b(N) \leq (b-1)c_b(N),$$

it is equivalent, up to constants depending only on b , to study c_b instead of s_b . We first give elementary logarithmic lower bounds based on divisibility. These arguments handle powers in which the base and the integer being exponentiated share prime factors, and they also handle $n!$ and $\Lambda_n = \text{lcm}(1, \dots, n)$. Finally, we discuss Stewart's theorem, which treats the nonzero base- b digits of powers a^n whenever a and b are multiplicatively independent, but requires Baker's theory of linear forms in logarithms.

Throughout, $a, b \geq 2$ are integers. For a prime p , $\nu_p(N)$ denotes the p -adic valuation of N .

2. A DIVISIBILITY ARGUMENT FOR POWERS

The following simple lemma is the basic mechanism. It says that sufficiently large prime-power divisibility forces nonzero digits to occur in many disjoint blocks.

Lemma 1. *Let $2 \leq a < b$ with $a \mid b$. Let $(e_k)_{k \geq 1}$ be integers such that $e_1 \geq 1$ and $a^{e_k} > b^{e_{k-1}}$ for $k \geq 2$. If N is divisible by a^{e_k} but not by b , then $c_b(N) \geq k$.*

Proof. The proof is by induction on k . For $k = 1$ the claim is immediate. For $k \geq 2$, write

$$N = b^{e_{k-1}}q + r, \quad 0 \leq r < b^{e_{k-1}}.$$

Since $N \geq a^{e_k} > b^{e_{k-1}}$, we have $q \geq 1$. Also r is divisible by $a^{e_{k-1}}$, because both N and $b^{e_{k-1}}q$ are; and r is not divisible by b , because N is not. Hence, by induction, $c_b(r) \geq k-1$. The base- b

expansion of N is obtained by adjoining the block of e_{k-1} low-order digits represented by r to the expansion of q , so

$$c_b(N) = c_b(q) + c_b(r) \geq 1 + (k - 1) = k.$$

□

Choosing $e_1 = 1$ and $e_k = \lceil re_{k-1} \rceil$, where $r = \log b / \log a > 1$, gives $e_k < r^k / (r - 1)$. Lemma 1 therefore implies the following explicit form.

Corollary 1. *Let $2 \leq a < b$ with $a \mid b$. If N is divisible by a^n but not by b , then*

$$c_b(N) > C \log n$$

for all sufficiently large n , where $C > 0$ depends only on a and b . Moreover, any $C < 1 / \log(\log b / \log a)$ is admissible.

To apply this to powers a^n , one must remove the trailing zeros in base b . The next lemma ensures that, under a multiplicative independence hypothesis, some prime shared by a and b survives to high order after those trailing zeros are removed.

Lemma 2. *Let $a, b \geq 2$, and suppose $\log a / \log b$ is irrational. If*

$$a^n = b^m t,$$

with $t \in \mathbb{Z}_{\geq 1}$, then there is a prime $p \mid a$ and a constant $C > 0$, depending only on a and b , such that

$$\nu_p(t) \geq Cn.$$

Proof. The vectors $(\nu_p(a))_p$ and $(\nu_p(b))_p$ are linearly independent over $\mathbb{Q}$. Hence there are primes p, q with

$$\nu_p(a)\nu_q(b) - \nu_q(a)\nu_p(b) > 0.$$

Comparing p -adic and q -adic valuations in $a^n = b^m t$ gives

$$n\nu_p(a) = m\nu_p(b) + \nu_p(t), \quad n\nu_q(a) \geq m\nu_q(b).$$

Eliminating m yields

$$\nu_p(t) \geq n \left(\nu_p(a) - \frac{\nu_q(a)}{\nu_q(b)} \nu_p(b) \right),$$

and the coefficient is positive by the choice of p, q . □

Let d be the part of a supported on the primes dividing b : equivalently, d is the smallest divisor of a such that $\gcd(a/d, b) = 1$. Combining Corollary 1 with Lemma 2 gives an elementary logarithmic lower bound in this shared-prime case.

Theorem 1. *Let $a, b \geq 2$, let d be the smallest divisor of a such that $\gcd(a/d, b) = 1$, and suppose that $\log d / \log b$ is irrational. Then*

$$c_b(a^n) > C \log n$$

for all sufficiently large n , with $C > 0$ depending only on a and b .

Proof. Write $a = dg$ with $\gcd(g, b) = 1$, and write $a^n = b^m s$ with $b \nmid s$. Since $g^n \mid s$, put $s = g^n t$. Then $d^n = b^m t$. By Lemma 2, there is a prime $p \mid d$ with $\nu_p(t) \geq C_1 n$. Since $t \mid s$, we also have $\nu_p(s) \geq C_1 n$. Applying Corollary 1 to s gives $c_b(s) > C \log n$. Finally, a^n and s differ only by a power of b , so they have the same nonzero base- b digits. □

This method cannot prove the full theorem of Senge–Straus and Stewart. For example, it treats multiples of powers; but every 3^n has a multiple of the form $10^k + 8$, which has only two nonzero decimal digits. A different idea is needed for powers such as 3^n in base 10.

3. FACTORIALS AND LEAST COMMON MULTIPLES

For factorials and least common multiples, the useful divisibility is not by a large prime power but by numbers of the form $b^r - 1$.

Lemma 3 (Balog–Dartyge[1], Stolarsky[8]). *If $m, r \geq 1$ and $b \geq 2$, and if $b^r - 1 \mid m$, then*

$$s_b(m) \geq (b - 1)r.$$

Proof. Group the base- b expansion of m into blocks of length r :

$$m = \sum_{i=0}^{k-1} B_i b^{ri}, \quad 0 \leq B_i < b^r.$$

Let $G(m) = \sum_i B_i$. Since $b^r \equiv 1 \pmod{b^r - 1}$, we have $G(m) \equiv m \pmod{b^r - 1}$. Moreover $G(m) < m$ whenever $m \geq b^r$, while $G(m) = m$ for $m < b^r$. Iterating G therefore carries any positive multiple of $b^r - 1$ down to the unique such multiple below b^r , namely $b^r - 1$. Subadditivity of digital sums gives

$$s_b(G(m)) \leq \sum_i s_b(B_i) = s_b(m),$$

so $s_b(m) \geq s_b(b^r - 1) = (b - 1)r$. □

Taking $m = n!$ or $m = \Lambda_n = \text{lcm}(1, \dots, n)$, we obtain a logarithmic lower bound immediately. If $n \geq b^r - 1$, then $b^r - 1$ divides both $n!$ and Λ_n . With $r = \lfloor \log_b(n + 1) \rfloor$, Lemma 3 yields

$$s_b(n!) \geq (b - 1) \lfloor \log_b(n + 1) \rfloor, \quad s_b(\Lambda_n) \geq (b - 1) \lfloor \log_b(n + 1) \rfloor.$$

Thus both digital sums are $\gg_b \log n$. Similar arguments appear in work of Luca [3]; Sanna [4], building on this lemma and deeper methods, proved the stronger bound

$$s_b(n!) > C_b \log n \log \log \log n$$

for $n > e^e$, and the same estimate holds for $s_b(\Lambda_n)$.

Heuristically, after ignoring trailing zeros, the remaining digits of $n!$ and Λ_n should be roughly equidistributed among $\{0, 1, \dots, b - 1\}$. This suggests much larger orders of growth, namely $s_b(n!) \asymp n \log n$ and $s_b(\Lambda_n) \asymp n$, but such estimates are presently out of reach.

4. STEWART'S THEOREM

The general result for powers requires lower bounds for linear forms in logarithms. We apply the following theorem, which we state without proof.

Theorem 2 (Baker–Wüstholz[2]). *Let*

$$\Lambda = b_1 \log \alpha_1 + \dots + b_n \log \alpha_n,$$

where $b_1, \dots, b_n$ are integers. Assume that $\alpha_1, \dots, \alpha_n$ are algebraic numbers with heights at most $A_1, \dots, A_n$ (all $\geq e$) respectively and that their logarithms have their principal values. Further assume that $b_1, \dots, b_n$ have absolute values at most B ($\geq e$). If $\Lambda \neq 0$ then

$$\log |\Lambda| > -(16nd)^{2(n+2)} \log A_1 \dots \log A_n \log B,$$

where d denotes the degree of $\mathbb{Q}(\alpha_1, \dots, \alpha_n)$.

We first record a small consequence of Lemma 2.

Lemma 4. *Let $a, b \geq 2$ and suppose $\log a / \log b$ is irrational. If $a^n = b^r t$ with t not divisible by b , then*

$$\log t \asymp_{a,b} n.$$

Proof. The upper bound follows from $t \leq a^n$. The lower bound follows from Lemma 2, since t is divisible by $p^{\lceil Cn \rceil}$ for some prime p . $\square$

Theorem 3 (Stewart[7]). *Let $a, b \geq 2$, and suppose $\log a / \log b$ is irrational. Then there is a constant C , depending only on a and b , such that*

$$c_b(a^n) > \frac{\log n}{\log \log n + C}$$

for all sufficiently large n .

Proof. Let $N = a^n$, and write its nonzero base- b digits in the form

$$N = b^m (d_1 b^{-m_1} + \dots + d_k b^{-m_k}),$$

where

$$1 = m_1 < m_2 < \dots < m_k \leq m, \quad 1 \leq d_j \leq b-1.$$

We shall prove that consecutive nonzero digit positions cannot be separated by too large a multiplicative factor. Fix $i < k$. Split N at the i -th nonzero digit:

$$q = b^{m_i} \sum_{j \leq i} d_j b^{-m_j}, \quad r = b^m \sum_{j > i} d_j b^{-m_j}.$$

Then q and r are positive integers and

$$N = b^{m-m_i} q + r.$$

Moreover,

$$b^{m_i-1} \leq q < b^{m_i},$$

because q is the integer formed by the first m_i digits of N , and its leading digit is nonzero. Similarly, since the first nonzero digit of r occurs at exponent $m - m_{i+1}$, we have

$$b^{m-m_{i+1}} \leq r < b^{m-m_{i+1}+1}.$$

Since $b^{m-1} < N < b^m$, it follows that

$$b^{-m_{i+1}} < \frac{r}{N} < b^{-m_{i+1}+2}.$$

Now define the logarithmic form

$$\Lambda = -n \log a + (m - m_i) \log b + \log q.$$

Then

$$e^\Lambda = a^{-n} b^{m-m_i} q = 1 - \frac{r}{N}.$$

The linear form Λ is nonzero. Indeed, if $\Lambda = 0$, then $N = b^{m-m_i} q$, which would force $r = 0$, contradicting the fact that there is at least one nonzero digit after the i -th one.

We apply Baker–Wüstholz to

$$\Lambda = -n \log a + (m - m_i) \log b + \log q.$$

The algebraic numbers are a , b , and q , with integer coefficients $-n$, $m - m_i$, and 1. The numbers a and b are fixed, while q varies. Since q is a positive integer, its height factor is $\asymp \max\{1, \log q\}$. Also $m = O_{a,b}(n)$, so the coefficient bound B in Baker–Wüstholz may be taken to satisfy $B = O_{a,b}(n)$. Hence

$$\log |\Lambda| > -C_1 \max\{1, \log q\} \log n.$$

If $m_i \geq 3$, then $q \geq b^{m_i-1} \geq b^2$, so in particular $\log q > 1$, and the preceding estimate becomes

$$\log |\Lambda| > -C_2 \log q \log n.$$

Using $q < b^{m_i}$, this gives

$$\log |\Lambda| > -C_3 m_i \log n.$$

On the other hand,

$$e^\Lambda = 1 - \frac{r}{N}.$$

Since $r/N < 1/2$, we have

$$|\Lambda| = -\log \left(1 - \frac{r}{N}\right) < 2\frac{r}{N}.$$

Using $r/N < b^{-m_{i+1}+2}$, we obtain

$$\log |\Lambda| < \log 2 + (-m_{i+1} + 2) \log b.$$

Combining the lower and upper bounds for $\log |\Lambda|$ yields

$$-C_3 m_i \log n < \log 2 + (-m_{i+1} + 2) \log b,$$

and hence

$$m_{i+1} \leq C_4 m_i \log n.$$

This proves the desired estimate when $m_i \geq 3$.

If $m_i < 3$, then q is bounded in terms of b . In this case the Baker–Wüstholz lower bound gives simply

$$\log |\Lambda| > -C_5 \log n.$$

The same upper bound

$$\log |\Lambda| < \log 2 + (-m_{i+1} + 2) \log b$$

then implies

$$m_{i+1} \leq C_6 \log n.$$

Since $m_i \geq 1$, this again gives

$$m_{i+1} \leq C_7 m_i \log n.$$

Thus for every $i < k$,

$$\frac{m_{i+1}}{m_i} \leq C_7 \log n.$$

Taking logarithms gives

$$\log m_{i+1} - \log m_i \leq \log \log n + C_8.$$

Summing over $i = 1, \dots, k-1$, and using $m_1 = 1$, we get

$$\log m_k \leq (k-1)(\log \log n + C_8).$$

It remains to relate m_k to n . Let T be the number of trailing zeroes in the base- b expansion of $N = a^n$, and write

$$N = b^T t, \quad b \nmid t.$$

The last nonzero digit of N occurs at exponent T . Since that same digit has exponent $m - m_k$, we have

$$T = m - m_k.$$

Therefore $t = N/b^T$ has exactly m_k base- b digits. Equivalently,

$$b^{m_k-1} \leq t < b^{m_k},$$

so

$$m_k = \frac{\log t}{\log b} + O(1).$$

By Lemma 4,

$$\log t \asymp_{a,b} n.$$

Hence

$$m_k \asymp_{a,b} n,$$

and therefore

$$\log m_k = \log n + O_{a,b}(1).$$

Combining this with the previous inequality gives

$$\log n + O_{a,b}(1) \leq (k-1)(\log \log n + C_8).$$

After increasing the constant in the denominator, we obtain

$$k > \frac{\log n}{\log \log n + C}$$

for all sufficiently large n . Since $k = c_b(a^n)$, this proves the theorem. $\square$

The irrationality hypothesis is necessary: if $\log a / \log b = r/s \in \mathbb{Q}$, then $a^{sn} = b^{rn}$ has only one nonzero base- b digit for every n . Thus Stewart's theorem supplies the quantitative form of the Senge–Straus criterion [5]: the nonzero digit count of a^n in base b tends to infinity exactly when a and b are multiplicatively independent.

REFERENCES

- [1] A. Balog and C. Dartyge, “On the sum of the digits of multiples,” *Moscow Journal of Combinatorics and Number Theory* **2** (2012), 3–15.
- [2] A. Baker and G. Wüstholz, “Logarithmic forms and group varieties,” *Journal für die reine und angewandte Mathematik* **442** (1993), 19–62. <https://doi.org/10.1515/crll.1993.442.19>
- [3] F. Luca, “The number of non-zero digits of $n!$,” *Canadian Mathematical Bulletin* **45** (2002), 115–118.
- [4] C. Sanna, “On the sum of digits of the factorial,” *Journal of Number Theory* **147** (2015), 836–841.
- [5] H. G. Senge and E. G. Straus, “PV-numbers and sets of multiplicity,” *Periodica Mathematica Hungarica* **3** (1973), 93–100.
- [6] W. Sierpiński, *250 Problems in Elementary Number Theory*, American Elsevier, 1970.
- [7] C. L. Stewart, “On the representation of an integer in two different bases,” *Journal für die reine und angewandte Mathematik* **319** (1980), 63–72.
- [8] K. B. Stolarsky, “Integers whose multiples have anomalous digital frequencies,” *Acta Arithmetica* **38** (1980), 117–128.